CAPSTONE PROJECT — FINAL REPORT

End-to-End Data Platform

Ingest → Lakehouse → Data Quality → BI → Governance

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1. EXECUTIVE SUMMARY

This project implements an end-to-end cloud data platform on Amazon Web Services (AWS).

The platform receives raw business order data, stores it in an Amazon S3 raw zone, automatically triggers an AWS Glue processing workflow when new files arrive, validates and transforms the data, and publishes trusted data into an Apache Iceberg curated data layer.

The curated data is queried using Amazon Athena and visualized using Amazon QuickSight. Data quality controls prevent invalid data from reaching the curated layer. Amazon Lake Formation provides fine-grained row-level access control, while Amazon CloudWatch, Amazon EventBridge and Amazon SNS provide monitoring and failure alerting.

The final platform demonstrates the complete trust chain:

Source → Raw S3 → EventBridge → Glue Workflow → Glue ETL → Data Quality → Curated Iceberg → Athena → QuickSight

The solution also demonstrates Iceberg time travel, snapshot history, data quality failure handling, quarantine of bad batches, row-level security, monitoring and dashboard reconciliation.

2. PROJECT OBJECTIVES

The main objectives of this project were:

Build an automated data ingestion pipeline.

Store raw data in Amazon S3.

Catalog raw data using AWS Glue.

Query raw data using Amazon Athena.

Build a curated data lakehouse using Apache Iceberg.

Partition and compress curated data for efficient analytics.

Implement data quality checks.

Prevent bad data from reaching downstream consumers.

Quarantine failed data batches.

Provide analytics using Amazon Athena and Amazon QuickSight.

Reconcile dashboard numbers with the underlying data.

Implement fine-grained access control using AWS Lake Formation.

Implement monitoring and alerting.

Demonstrate Iceberg time travel and snapshot history.

Document the complete architecture and operational process.

3. TECHNOLOGY STACK

The following AWS and open-source technologies were used:

Amazon S3

AWS Glue

AWS Glue Data Catalog

AWS Glue ETL

AWS Glue Workflow

Apache Iceberg

Amazon Athena

Amazon QuickSight

AWS Lake Formation

Amazon EventBridge

Amazon SNS

Amazon CloudWatch

Amazon Data Quality / Glue Data Quality

Apache Parquet

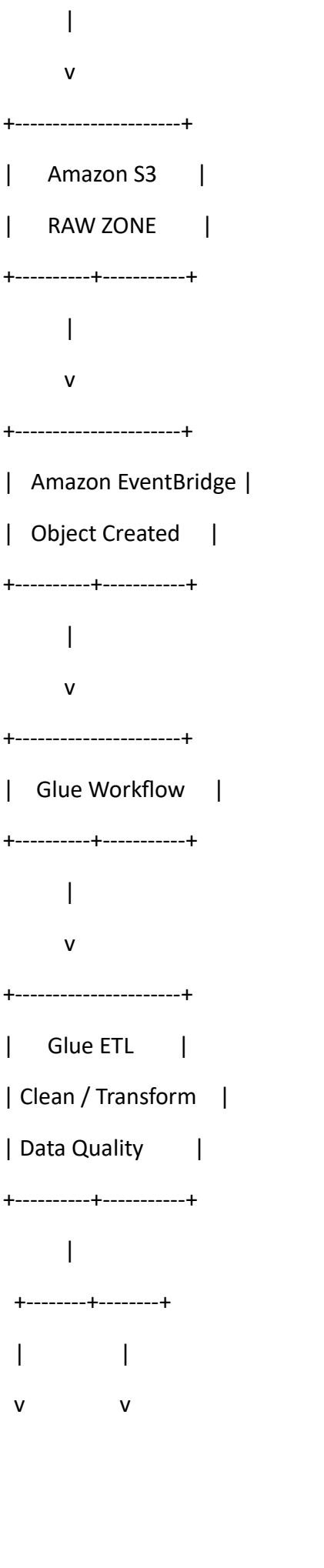
SQL

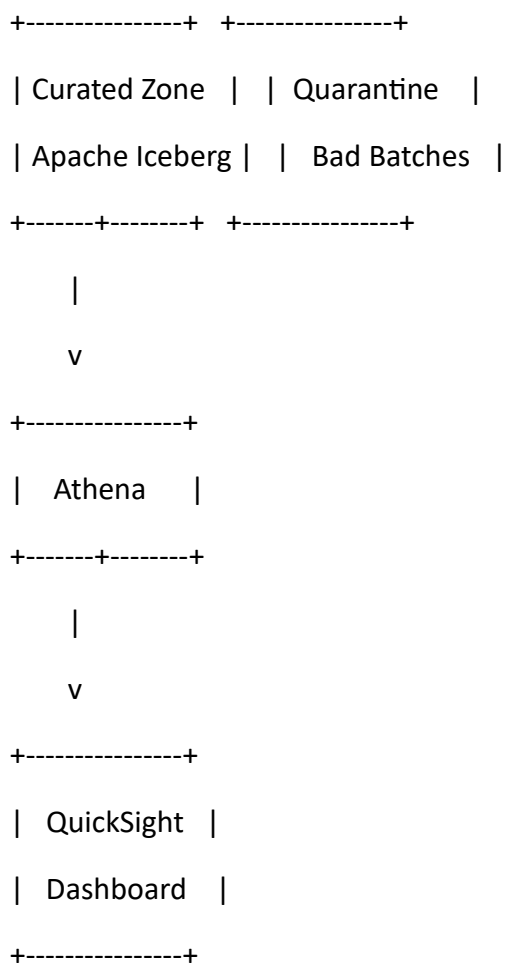
Python / PySpark

4. HIGH-LEVEL ARCHITECTURE

The platform follows a layered data architecture.

SOURCE DATA


Amazon S3
RAW ZONE
Amazon EventBridge
Object Created
Glue Workflow
Glue ETL
Clean / Transform
Data Quality



Lake Formation → Fine-Grained Access Control

CloudWatch → Monitoring / DQ Observability

EventBridge → SNS → Failure Alerts

5. AMAZON S3 DATA LAKE STRUCTURE

The main S3 bucket used in the project is:

shreik-capstone-sahil-2026

The logical structure is:

s3://shreik-capstone-sahil-2026/

raw/

orders/

streaming/

reference/

curated/

orders/

good_orders/

quarantine/

orders/

Raw Zone

The raw zone stores incoming source files without making them the final trusted analytical layer.

Curated Zone

The curated zone contains cleaned and transformed data stored using Apache Iceberg.

Quarantine Zone

The quarantine zone is used for data batches that fail data quality validation.

6. PHASE 1 — INGESTION AND RAW DATA ZONE

6.1 Raw Data Ingestion

The project starts with raw order data stored in Amazon S3.

The primary raw dataset contains:

200,250 rows

The raw data contains realistic data quality issues so that the platform can demonstrate validation and cleaning.

6.2 AWS Glue Data Catalog

AWS Glue Crawler was used to discover and catalog the raw order data.

Glue database:

shreik_capstone_db

Raw crawler:

shreik-raw-orders-crawler

Raw table:

raw_raw_orders

S3 source:

s3://shreik-capstone-sahil-2026/raw/orders/

7. RAW DATA ATHENA BASELINE

The raw data was queried using Amazon Athena.

Example query:

```
SELECT COUNT(*) AS total_rows
FROM raw_raw_orders;
```

Result:

200,250 rows

A regional aggregation was also performed:

```
SELECT
  region,
  COUNT(*) AS order_count,
  ROUND(SUM(net_amount), 2) AS total_sales
FROM raw_raw_orders
GROUP BY region
ORDER BY region;
```

The raw query baseline showed:

Data scanned: 26.94 MB

This baseline was later compared with the curated Iceberg layer.

8. AUTOMATED INGESTION TRIGGER

Amazon EventBridge was configured to detect new objects arriving in the raw orders S3 prefix.

EventBridge rule:

shreik-capstone-s3-to-glue

Event pattern:

S3 Object Created

↓

raw/orders/

↓

EventBridge

↓

Glue Workflow

The Glue workflow:

shreik-capstone-orders-workflow

The workflow triggers:

shreik-capstone-orders-etl

The automated pipeline was successfully tested by uploading a new order file to the raw S3 prefix.

The EventBridge monitoring showed matched events and successful invocations, followed by successful Glue workflow execution.

Therefore, the automated ingestion trigger was successfully demonstrated.

9. STREAMING DATA PATH

A small streaming ingestion path was also implemented using Amazon Kinesis Data Firehose.

Firehose:

shreik-capstone-firehose

Destination:

s3://shreik-capstone-sahil-2026/raw/streaming/

The Firehose configuration uses Direct PUT as the ingestion method and delivers streaming-style event data into S3.

This provides a lightweight streaming ingestion demonstration alongside the batch ingestion pipeline.

10. PHASE 2 — ORCHESTRATED ETL AND LAKEHOUSE

10.1 Glue Workflow

The ETL pipeline was orchestrated using AWS Glue Workflow.

Workflow:

shreik-capstone-orders-workflow

Workflow structure:

S3 Orders Arrival

↓

s3-orders-arrival-trigger

↓

shreik-capstone-orders-etl

The workflow successfully completed both manually and through S3-triggered execution.

11. GLUE ETL PROCESS

The Glue ETL job performs the following major operations:

Read source data.

Validate fields.

Clean invalid records.

Convert data into an analytical format.

Generate order date.

Write the curated data.

Store the curated data as Apache Iceberg.

Glue job:

shreik-capstone-orders-etl

Glue version:

AWS Glue 5.0

Worker configuration:

G.1X

Number of workers:

2

12. APACHE ICEBERG CURATED TABLE

The curated table is:

curated_orders

Database:

shreik_capstone_db

The table uses Apache Iceberg.

Important properties:

Apache Iceberg table

Daily partitioning

ZSTD compression

Athena queryable

Snapshot history

Time travel capability

Partitioning:

day(order_date)

Location:

s3://shreik-capstone-sahil-2026/curated/orders

13. CURATED TABLE SCHEMA

The main columns include:

order_id

order_ts

customer_id

product_id

category

region

quantity

unit_price

discount_pct

payment_method

status

shipping_days

gross_amount

discount_amount

net_amount

order_date

The order_date column is used for daily partitioning.

14. CURATED DATA RESULT

After transformation and quality filtering, the curated Iceberg table contained:

162,164 rows

Athena query:

SELECT COUNT(*) AS total_rows

FROM curated_orders;

Result:

162,164

15. RAW VS CURATED PERFORMANCE / SCAN COMPARISON

The same regional aggregation was executed on both raw and curated data.

Raw

Data scanned:

26.94 MB

Curated

Data scanned:

662.04 KB

The curated query scanned approximately:

40.7× less data

This demonstrates the storage and analytical efficiency benefits of the curated partitioned and compressed data layer.

Important observation:

The particular test execution did not show a lower query runtime for the curated query. Therefore, the measured improvement is reported as data scanned reduction, rather than claiming a query latency improvement.

16. ICEBERG TIME TRAVEL

Apache Iceberg time travel was successfully demonstrated using Athena Engine 3.

Query:

```
SELECT COUNT(*) AS historical_rows

FROM curated_orders

FOR TIMESTAMP AS OF

(current_timestamp - interval '1' day);
```

Result:

162,164 historical rows

This proves that historical versions of the Iceberg table can be queried using time travel.

17. ICEBERG SNAPSHOT HISTORY

The Iceberg snapshot metadata table was queried using:

```
SELECT

  snapshot_id,

  committed_at,

  operation

FROM "shreik_capstone_db"."curated_orders$snapshots"

ORDER BY committed_at DESC;
```

The query returned:

4 snapshots

The snapshots included multiple commit timestamps and overwrite operations.

This demonstrates that the Iceberg table maintains snapshot history.

18. PHASE 3 — DATA QUALITY GATES

Data quality is a key part of the platform.

The quality checks cover:

Row counts

Null values

Quantity validation

Price validation

Region validation

Status validation

Business rules

Valid order statuses are:

PLACED

SHIPPED

DELIVERED

CANCELLED

19. GOOD BATCH VALIDATION

A clean batch was processed through the data quality gate.

Results:

Total rows : 4133

Valid rows : 4133

Invalid rows : 0

DQ STATUS : PASSED

The valid batch was successfully published to:

s3://shreik-capstone-sahil-2026/curated/good_orders/

Athena verification confirmed:

4,133 curated rows

Therefore, the good batch successfully passed the data quality gate.

20. BAD BATCH VALIDATION

A bad batch was intentionally tested to prove the failure path.

Results:

Total rows : 5000

Valid rows : 4133

Invalid rows : 867

DQ STATUS : FAILED

The pipeline prevented the invalid batch from being published to the curated downstream layer.

The failed batch was quarantined.

This proves the important trust principle:

Bad data is stopped before reaching trusted downstream consumers.

21. DATA QUALITY OBSERVABILITY

AWS CloudWatch was used to monitor the data quality results.

The CloudWatch logs showed:

GOOD BATCH DATA QUALITY GATE

Total rows : 4133

Valid rows : 4133

Invalid rows : 0

DQ STATUS : PASSED

The bad batch logs showed:

Total rows : 5000

Valid rows : 4133

Invalid rows : 867

DQ STATUS : FAILED

This provides operational visibility into the data quality gate.

22. PHASE 4 — QUERY LAYER AND BUSINESS INTELLIGENCE

Amazon Athena provides the SQL query layer for the curated data.

Example query:

```
SELECT
  region,
```

```
    COUNT(*) AS order_count,
    ROUND(SUM(net_amount), 2) AS total_sales
FROM curated_orders
GROUP BY region
ORDER BY region;
```

Results:

Region

Order Count

Total Sales

Central

28,988

–

East

29,322

–

North

37,515

–

South

32,290

–

West

34,049

–

Total order count:

162,164

23. AMAZON QUICKSIGHT DASHBOARD

Amazon QuickSight was connected to the curated Athena dataset.

Dataset:

curated_orders

The dataset was configured using SPICE.

The dashboard contains:

Total Orders KPI

Total Net Sales KPI

Regional sales visualization

Region filter

Interactive business analysis

24. QUICKSIGHT RESULTS

The dashboard showed:

Total Orders: 162,164

Total Net Sales: 62,441,156.12

The dashboard was tested using the Region filter.

For example, selecting North changed the dashboard KPIs to the North-region values.

Removing the filter restored the overall totals.

25. ATHENA AND QUICKSIGHT RECONCILIATION

The dashboard numbers were reconciled with Athena.

Athena:

Total Orders = 162,164

QuickSight:

Total Orders = 162,164

Athena / QuickSight sales:

Total Net Sales = 62,441,156.12

Therefore, the dashboard values reconcile with the curated analytical data.

This establishes the trust chain:

Curated Iceberg

↓

Athena

↓

QuickSight

26. QUICKSIGHT REFRESH

The QuickSight dataset was configured with a scheduled refresh.

A manual refresh was also successfully tested.

Latest successful refresh evidence showed:

Status : Completed

Rows ingested: 162,164

Dataset rows : 162,164

The refresh duration observed during testing was approximately:

2 minutes 25 seconds

27. PHASE 5 — GOVERNANCE AND SECURITY

AWS Lake Formation was used to implement fine-grained data access.

The main catalog database is:

shreik_capstone_db

The S3 data lake location was registered with Lake Formation.

28. ROW-LEVEL SECURITY

Two analyst roles were configured.

North Analyst

IAM role:

Capstone-Analyst-North

Lake Formation filter:

North_Orders_Filter

The filter restricts the analyst to North region records.

West Analyst

IAM role:

Capstone-Analyst-West

Lake Formation filter:

west_orders_filter

The filter restricts the analyst to West region records.

Therefore, the same curated table can provide different row-level views to different users.

29. DATA CATALOG AND METADATA

The curated table description was documented in AWS Glue.

Table:

curated_orders

Description:

Curated Iceberg orders table for trusted sales analytics and QuickSight reporting.

The table metadata identifies the curated dataset and its role in the analytics platform.

Business ownership and glossary information can additionally be maintained in the project documentation.

30. LINEAGE AND IMPACT ANALYSIS

The important data lineage is:

Raw Orders

↓

Glue ETL

↓

Curated Iceberg

↓

Athena

↓

QuickSight

Important column-level relationships include:

region

Source region

↓

curated_orders.region

↓

QuickSight Region Filter

↓

Regional Chart

net_amount

Source net_amount

↓

curated_orders.net_amount

↓

Athena Sales Aggregation

↓

QuickSight Total Net Sales

order_id

Source order_id

↓

curated_orders.order_id

↓

Order Count / Record Identification

If the region column changes or is removed, the QuickSight Region filter and regional chart are impacted.

If the net_amount column changes or is removed, the Total Net Sales KPI and regional sales analysis are impacted.

31. CLOUDWATCH OBSERVABILITY

A CloudWatch dashboard was created:

shreik-capstone-observability

The dashboard provides visibility into:

Data quality results

Good batch status

Bad batch status

Glue processing logs

Pipeline operational information

Data freshness SLA documentation

A CloudWatch Logs Insights query was used to search for DQ events.

Example:

SOURCE logGroups(namePrefix: [], class: "STANDARD") START=-1w END=0s

| fields @timestamp, @message

| filter @message like /DQ_STATUS|DATA_QUALITY|GOOD_BATCH/

| sort @timestamp desc

| limit 50

32. DATA FRESHNESS SLA

The project documents a freshness SLA for the curated dataset.

Data Freshness SLA

Curated orders data should be available within 24 hours of source ingestion.

Monitoring:

AWS CloudWatch

Quality monitoring:

Good and bad batch DQ results

Failure alerting:

AWS Glue FAILED

↓

Amazon EventBridge

↓

Amazon SNS

The SLA is documented and monitored operationally. The project does not claim a separate automated 24-hour threshold alarm unless explicitly configured.

33. FAILURE ALERTING

An EventBridge rule was created to detect Glue job failures.

Rule:

shreik-capstone-glue-failure-alerts

Event:

Glue Job State Change

Failure state:

FAILED

Job:

shreik-capstone-orders-etl

Target:

Amazon SNS

SNS topic:

Shreik-capstone-alerts

Architecture:

Glue ETL

|

| FAILED

v

EventBridge

|

v

SNS

|

v

Alert / Notification

This provides automated failure alerting for the Glue ETL process.

34. END-TO-END TRUST CHAIN

The complete trust chain is:



This architecture ensures that invalid data is prevented from reaching trusted analytical consumers.

35. SECURITY MODEL

The project applies multiple layers of security:

IAM roles for AWS service execution.

S3 access control.

Lake Formation data lake registration.

Lake Formation row-level filters.

Separate analyst roles.

Controlled Athena access.

QuickSight dataset permissions.

The principle of least privilege is applied where practical through service-specific IAM roles and Lake Formation filters.

36. COST AND EFFICIENCY

The curated layer significantly reduces Athena data scanning.

Measured comparison:

Raw scan = 26.94 MB

Curated scan = 662.04 KB

Reduction:

Approximately 40.7× less data scanned

This can reduce Athena query cost because Athena pricing is based on data scanned.

The use of:

Partitioning

Parquet

Compression

Curated data

Iceberg

helps improve analytical storage and query efficiency.

37. KEY PROJECT METRICS

Metric

Result

Raw dataset

200,250 rows

Curated dataset

162,164 rows

Raw query scan

26.94 MB

Curated query scan

662.04 KB

Scan reduction

~40.7×

Good batch

4,133 rows

Good batch invalid rows

0

Bad batch

5,000 rows

Bad batch invalid rows

867

Iceberg snapshots

4

QuickSight orders

162,164

QuickSight net sales

62,441,156.12

Freshness SLA

24 hours

38. LIVE DEMONSTRATION PLAN

The final live demonstration can be performed in the following order.

Step 1 — Show S3

Open:

Amazon S3

→ shreik-capstone-sahil-2026

→ raw

→ orders

Explain that this is the raw landing zone.

Step 2 — Upload New Data

Upload a new order file.

Explain:

New file arrival automatically triggers the pipeline.

Step 3 — Show EventBridge

Open:

EventBridge

→ Rules

→ shreik-capstone-s3-to-glue

Show that the rule detects the S3 Object Created event.

Step 4 — Show Glue Workflow

Open:

AWS Glue

→ Workflows

→ shreik-capstone-orders-workflow

Show:

S3 Event

↓

Glue ETL

Show the successful workflow run.

39. BAD BATCH LIVE DEMO

Upload or process the bad batch.

Show the CloudWatch / Glue output:

Total rows : 5000

Valid rows : 4133

Invalid rows : 867

DQ STATUS : FAILED

Explain:

The quality gate prevents bad data from being published to the trusted curated layer.

Then show the quarantine location.

40. ATHENA RECONCILIATION DEMO

Run:

```
SELECT COUNT(*) AS total_rows
FROM curated_orders;
```

Expected result:

162164

Then run:

```
SELECT
  region,
  COUNT(*) AS order_count,
  ROUND(SUM(net_amount), 2) AS total_sales
FROM curated_orders
GROUP BY region
ORDER BY region;
```

Show the regional results.

41. QUICKSIGHT DEMO

Open the QuickSight dashboard.

Show:

Total Orders

162,164

Show:

Total Net Sales

62,441,156.12

Apply the Region filter.

Show that the dashboard changes according to the selected region.

Remove the filter and show that the complete totals return.

42. LAKE FORMATION DEMO

Show the configured filters:

North_Orders_Filter

and

west_orders_filter

Explain:

Different analyst roles can receive different rows from the same curated table.

43. ICEBERG DEMO

Show:

```
SELECT COUNT(*) AS historical_rows
FROM curated_orders
FOR TIMESTAMP AS OF
(current_timestamp - interval '1' day);
```

Expected result:

162164

Then show the snapshot history:

```
SELECT
   snapshot_id,
   committed_at,
   operation
FROM "shreik_capstone_db"."curated_orders$snapshots"
ORDER BY committed_at DESC;
```

Expected:

4 snapshots

44. MONITORING DEMO

Open:

CloudWatch

→ Dashboards

→ shreik-capstone-observability

Show:

Good batch passed

Bad batch failed

Invalid row count

Processing logs

Freshness SLA

Then explain the alerting path:

Glue Failure

↓

EventBridge

↓

SNS

45. PROJECT VALIDATION

The project has been validated across all major layers.

Ingestion

New S3 objects trigger the automated workflow.

Processing

Glue ETL processes and transforms the data.

Data Quality

Valid batches pass and invalid batches fail.

Storage

Curated data is stored using Apache Iceberg.

Query

Athena successfully queries the curated data.

BI

QuickSight successfully visualizes the curated data.

Reconciliation

QuickSight totals match Athena totals.

Governance

Lake Formation provides row-level filtering.

Monitoring

CloudWatch provides operational visibility.

Alerting

EventBridge and SNS provide Glue failure alerts.

46. CHALLENGES AND SOLUTIONS

Challenge 1 — Glue Spark configuration

An initial Glue configuration produced a static Spark configuration error.

The configuration was corrected to allow the Iceberg ETL job to run successfully.

Challenge 2 — S3 permissions

The Glue ETL job initially encountered an S3 PutObject permission issue.

The required IAM/S3 permissions were corrected.

The ETL job then successfully wrote the curated Iceberg data.

Challenge 3 — QuickSight S3 permissions

QuickSight initially could not access the required S3 resources for dataset refresh.

The required S3 permission was enabled in QuickSight resource permissions.

The dataset refresh then completed successfully.

Challenge 4 — Data quality

The source data intentionally contained invalid records.

Instead of publishing these records directly, a quality gate was implemented.

Invalid batches are rejected and quarantined.

47. LESSONS LEARNED

This project demonstrated the importance of separating:

Raw data

Processing

Data quality

Curated data

Analytics

Governance

Monitoring

The project also demonstrated that data quality should be treated as a pipeline gate rather than as a downstream reporting problem.

Apache Iceberg provides useful capabilities such as:

Snapshot history

Time travel

Schema evolution support

Transactional table management

Lake Formation provides fine-grained data access controls, while CloudWatch and EventBridge improve operational visibility.

48. PROJECT OUTCOME

The project successfully demonstrates an end-to-end AWS data platform.

The final solution provides:

Automated ingestion

+

Data transformation

+

Data quality

+

Lakehouse storage

+

SQL analytics

+

BI dashboard

+

Governance

+

Monitoring

+

Alerting

The platform converts raw incoming data into trusted business information while preventing failed data batches from reaching downstream consumers.

49. CONCLUSION

The End-to-End Data Platform project successfully implements a cloud-based data engineering architecture using AWS services.

The platform starts with raw data ingestion into Amazon S3 and continues through automated orchestration, data transformation, data quality validation, Apache Iceberg storage, Athena analytics and QuickSight business intelligence.

The implementation also includes governance through Lake Formation, operational monitoring through CloudWatch, and automated Glue failure alerting through EventBridge and SNS.

The final dashboard is reconciled with the underlying curated data, demonstrating a complete trust chain from source data to business reporting.

The project therefore demonstrates the practical implementation of a modern cloud data platform with ingestion, lakehouse processing, data quality, analytics, governance, observability and business intelligence.

50. APPENDIX — IMPORTANT AWS RESOURCE NAMES

S3 Bucket:

shreik-capstone-sahil-2026

Glue Database:

shreik_capstone_db

Glue Raw Crawler:

shreik-raw-orders-crawler

Glue ETL:

shreik-capstone-orders-etl

Glue Workflow:

shreik-capstone-orders-workflow

EventBridge S3 Rule:

shreik-capstone-s3-to-glue

EventBridge Failure Rule:

shreik-capstone-glue-failure-alerts

SNS Topic:

Shreik-capstone-alerts

Glue Curated Table:

curated_orders

Glue DQ Ruleset:

capstone_orders_dq_rules

IAM Role:

AWSGlueServiceRole-Capstone

North Analyst:

Capstone-Analyst-North

West Analyst:

Capstone-Analyst-West

Lake Formation Filter:

North_Orders_Filter

Lake Formation Filter:

west_orders_filter

CloudWatch Dashboard:

shreik-capstone-observability

QuickSight Dashboard:

Shreik Capstone Sales dashboard